Anchor-Constrained Feature-Similarity Ensembling for Multimodal Lifelog State Prediction

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Abstract—Predicting daily sleep and stress states from lifelogs is difficult when labels are scarce, observations are participant-dependent, and validation does not match the hidden test distribution. We present an anchor-constrained ensemble for seven binary targets in the 5th ETRI Human Understanding AI Paper Challenge. A stable sequence-based anchor is blended with a pre-existing multi-signal model carrying a small same-participant nearest-neighbor correction over four sensor families. On 450 labeled person-days from 10 participants, the complete lineage ensemble obtains a non-nested local mean log loss of 0.534000 and a DACON public score of 0.5960832519, improving the previous public anchor by 0.0053280490. An OOF ablation attributes only 0.000508 of the 0.004638 local gain to the similarity correction; no public ablation isolates its leaderboard effect. More aggressive models transferred poorly, and three later near-anchor probes did not improve the result. The contribution is therefore the conservative ensemble and its explicit attribution and risk audit, not evidence that feature similarity alone caused the public gain.

Index Terms—lifelog, multimodal sensing, sleep prediction, stress prediction, nearest neighbors, ensemble learning

I. Introduction

Continuous sensing from smartphones and wearables provides a nonintrusive view of activity, physiology, mobility, and device use. The ETRI lifelog datasets combine these streams with subjective reports and sleep-health outcomes, enabling models of daily human state [1], [2]. Their small number of participants and heterogeneous routines, however, make generalization unusually fragile: a pattern may represent a participant's stable habit rather than a transferable state marker.

The 5th ETRI Human Understanding AI Paper Challenge asks participants to predict seven daily binary targets from multimodal lifelog data [3]. Our experiments exposed a second difficulty beyond ordinary overfitting. A candidate with local mean log loss 0.378934 obtained a much worse public score of 0.6698787318, whereas several models around local 0.54 transferred near public 0.60. Consequently, selecting the local minimum was actively misleading.

We address this setting with an anchor-constrained probability ensemble. The model combines a stable sequence-based prediction with a multi-signal predecessor that receives a low-weight, same-participant feature-similarity correction. The final prediction is shrunk toward a previously public-scored anchor, limiting the effect of the locally selected source. Our contributions are: (1) a target-specific similarity correction over domain-aligned sensor families with an explicit predecessor ablation; (2) an anchor-shrink strategy that improved the public score while bounding prediction movement; and (3) an empirical audit that keeps post-hoc leaderboard and OOF-slice evidence distinct from model selection. Because earlier public scores informed the anchor and operating policy, these are leaderboard-adapted competition findings rather than an independent generalization estimate.

II. Data and Evaluation

A. Task and Features

The data contain 450 labeled and 250 test person-days from 10 participants. Training dates span June 4 to November 15, 2024; test dates span July 7 to November 20, 2024. Test days are interleaved with labeled days for each participant rather than forming a pure future holdout. Excluding identifiers, dates, and targets, the processed representation has 232 numeric features spanning screen and app use, phone activity and charging, pedometer and heart rate, phone/watch light, GPS and mobility, Wi-Fi, Bluetooth, ambience, sleep-window summaries, and demographic/calendar context.

Participant contributions are unequal: labeled days range from 33 to 57 and test days from 19 to 32 per participant. Positive-label prevalence ranges from 0.496 for Q1 to 0.682 for S1. This is large enough to make an unstratified random split sensitive to individual baselines. The hidden test geometry is also heterogeneous. Six participants have two contiguous test blocks, while four have five to seven fragmented blocks. We therefore preserve participant identity during validation and treat a purely chronological split only as a stress test.

The subjective targets are sleep quality (Q1), low physical fatigue before sleep (Q2), and low stress before sleep (Q3). The objective sleep targets represent guideline adherence for total sleep time (S1), sleep efficiency (S2), sleep onset latency (S3), and wake after sleep onset (S4). Following the competition metric, we use mean binary log loss over the seven targets:

$$\mathcal{L} = -\frac{1}{7N}\sum_{i=1}^{N}\sum_{t=1}^{7}\left[y_{it}\log p_{it} + (1-y_{it})\log(1-p_{it})\right]. \quad (1)$$

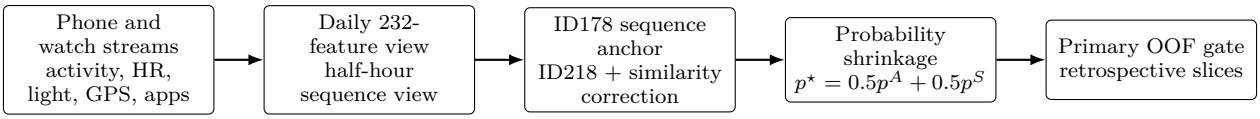


Fig. 1. Overview of the anchor-constrained workflow. Labels enter model training and same-participant neighbor sets. Public scores informed anchor choice and the retrospective transfer audit, but not parameter fitting.

Prior work has represented synchronized lifelog channels as images [4] and learned cross-modal time-series representations with latent masking [5]. Our limited sample size instead favored strong shrinkage and compact probability overlays. We retained raw sequence information through ROCKET-style random convolution features [6], but used sensor-family nearest neighbors for the final correction.

B. Validation Lenses

The primary local lens is five-fold, within-participant blocked validation designed to resemble the interleaved test pattern. After public scoring, we also summarized frozen OOF predictions over subject, temporal, and test-boundary slices. Historically named subject-hole and LOSO reports in the experiment archive are slice diagnostics: they do not refit the complete model and therefore do not estimate unseen-participant generalization. The public leaderboard is sparse post-hoc competition evidence, not a validation fold.

Primary validation folds contain 85–95 days and preserve observations from all participants where possible. Retrospective slices emphasize participant subsets, contiguous temporal regions, or labeled days adjacent to known test blocks. They answer descriptive stability questions and are not averaged into a synthetic score or presented as independent validation. Candidate selection uses the primary lens; the slices only audit where its frozen OOF errors concentrate.

Feature-similarity hyperparameters were selected on the full primary out-of-fold diagnostic, not by nested cross-validation. The grid compared 216 settings per target (four families, three neighbor counts, three temporal penalties, and six blend weights), or 1,512 target/configuration evaluations. The selected corrections improved only 21 of 35 target-fold losses relative to ID218 despite improving all seven aggregate target losses, indicating material fold dependence. Thus, 0.534000 is an optimistically biased model-selection score. Label-free median imputation and standardization used the combined train/test covariate matrix; all neighbor outcomes came only from labeled training rows.

III. Method

A. Stable Sequence Anchor

The public anchor, denoted p^A (experiment ID178), is a target-wise ensemble built on processed daily features and half-hour sequence summaries. Deterministic random convolution features are followed by regularized logistic heads. Depending on the target, 192 or 384 kernels are used and the sequence probability receives weight 0.11–0.165; S3 retains its preceding base prediction. Although the raw sequence heads alone overfit, their small probability overlays produced a stable public score of 0.6014113009. We use this scored model as an empirical anchor rather than claiming that its local score is unbiased.

B. Same-Participant Feature Similarity

We define four token-selected, potentially overlapping feature families: date/time (29 features), screen/app (53), movement/GPS (44), and light/heart rate (114). Counts are not additive because a feature may match more than one family. Missing values are median-imputed and each feature is standardized. For a query day i and a labeled day j belonging to the same participant, family-specific distance is

$$d_{ij}^{(f)} = \sqrt{\frac{1}{|f|}\|z_i^{(f)} - z_j^{(f)}\|_2^2} + \lambda \log(1 + |\tau_i - \tau_j|), \qquad (2)$$

where $z^{(f)}$ is the standardized feature vector and τ is calendar date. The similarity estimate s_{it} is the mean label among the k nearest eligible days of the same participant, with $k \in \{1, 3, 7\}$ and $\lambda \in \{0, 0.03, 0.1\}$. This follows the local-neighborhood intuition of nearest-neighbor classification [7] while preventing labels from other participants from defining an individual's baseline.

The similarity predecessor p^B is experiment ID218, a pre-existing multi-signal ensemble with local loss 0.534103. It is distinct from the public anchor and must be supplied to reproduce the correction. For each target, a small blend forms source ID221:

$$\begin{aligned} p_{it}^S &= (1 - w_t)p_{it}^B + w_t s_{it}, \\ w_t &\in \{0.005, 0.01, 0.02, 0.03, 0.05, 0.07\}. \end{aligned} \qquad (3)$$

Table I lists the selected target-specific configurations. The search remained deliberately narrow, and all overlay weights were at most 0.07. Thus, ID221 is an ID218-based corrected source, not a standalone nearest-neighbor model.

C. Anchor-Constrained Shrinkage and Promotion

The final model (ID342) is the midpoint between the stable public anchor and the similarity source:

$$p_{it}^\star = (1 - \alpha)p_{it}^A + \alpha p_{it}^S, \qquad \alpha = 0.50. \qquad (4)$$

This second shrinkage reduced the maximum test-probability movement relative to p^A from the unshrunk source while retaining half of its locally useful direction.

TABLE I
Selected feature-similarity configurations.

Target	Family	k	λ	w_t
Q1	screen/app	1	0	0.07
Q2	date/time	1	0.10	0.03
Q3	date/time	1	0	0.03
S1	date/time	1	0.03	0.02
S2	move/GPS	7	0.03	0.07
S3	screen/app	1	0.03	0.05
S4	light/HR	3	0.03	0.07

The resulting mean and maximum absolute test changes were 0.01846 and 0.09391.

The value $\alpha = 0.50$ was chosen once as a conservative midpoint before ID342 was publicly scored, not by an α leaderboard sweep. For $\alpha \in \{0, 0.25, 0.50, 0.75, 1\}$, primary OOF losses were 0.538638, 0.535813, 0.534000, 0.533074, and 0.532990. Although larger values improved the selected full-OOF mean, $\alpha = 0.50$ improved all five fold losses over ID178, versus four of five for 0.75 and 1. This sensitivity is selection-conditioned and does not make the midpoint independently validated.

Pre-submission promotion used three checks. First, target and mean losses had to improve under the primary lens. Second, probability movement was compared with two already scored failed directions: ID287, an extreme local frontier, and ID329, a guarded candidate that did not improve the public anchor. Third, every scored result was tied to a prediction artifact, experiment record, submission file, and official score. Subject and temporal OOF-slice analyses were conducted retrospectively and were not pre-submission promotion evidence. Failed-frontier rows were retained as negative controls.

For candidate c and anchor A, we recorded mean and maximum test movement, $m_1 = \frac{1}{7N}\sum_{it} |p^c_{it} - p^A_{it}|$ and $m_\infty = \max_{it} |p^c_{it} - p^A_{it}|$. These quantities do not estimate accuracy; they bound how much an unscored model can invalidate what is known about the anchor. We also measured cosine projection onto the ID287 and ID329 prediction differences. Promotion required small movement and no dominant alignment with either failed direction. The thresholds were operating safeguards fixed for the final slate, not tuned accuracy metrics.

IV. Results

Table II separates the ID218 predecessor from its ID221 similarity correction. The correction improves all seven target losses relative to ID218, most visibly Q1, but ID218 already supplies most of the difference from ID178. Midpoint shrinkage limits reliance on the non-nested source selection.

An OOF-only midpoint ablation, $0.5p^A + 0.5p^B$, scores 0.534508. Moving from this ablation to ID342 improves local loss by only 0.000508, or 11.0% of ID342's 0.004638 gain over ID178. ID218, ID221, and the midpoint ablation were not publicly scored, so the similarity correction's public contribution is unidentified. Table III supports only the complete-lineage comparison: ID342 improves ID178 by 0.0053280490 and the preceding best public score by 0.0045373767. ID287 demonstrates that unrestricted local-frontier selection can reverse the ranking.

TABLE II
Primary-lens log loss by target (lower is better).

Target	ID178	ID218	ID221	ID342
Q1	0.551916	0.548015	0.544061	0.545848
Q2	0.556679	0.550689	0.550171	0.551609
Q3	0.567090	0.562768	0.562231	0.563226
S1	0.471630	0.467245	0.467117	0.466643
S2	0.529556	0.529516	0.528881	0.529101
S3	0.490893	0.490545	0.489489	0.489945
S4	0.602700	0.589944	0.588980	0.591629
Mean	0.538638	0.534103	0.532990	0.534000

TABLE III
Public-scored models used as transfer controls.

ID	Local	Public	Role
342	0.534000	0.596083252	final model
339	0.536441	0.598518779	sequence summary
338	0.537861	0.599214518	random-conv edit
337	0.538066	0.600620629	low-movement edit
340	0.538426	0.601379175	Q2 sequence edit
329	0.516718	0.601790858	guarded failure
287	0.378934	0.669878732	frontier failure
178	0.538638	0.601411301	prior anchor

To quantify this effect retrospectively, Table IV compares local/public rank agreement under progressively stricter control sets. Pairwise agreement is the fraction of candidate pairs ordered identically by local and public score. Removing only ID287 is insufficient because ID329 remains locally attractive but publicly weaker than ID178. After excluding both public-identified failures, the six remaining models have exact rank agreement. The three-candidate June 23 slate also has the same ordering across post-hoc OOF slices. This is descriptive and selection-conditioned: the exclusions use observed public failures, so the perfect agreement neither independently validates the promotion rule nor constitutes prospective validation.

Three final Q3 app-name lexicon anti-signal variants provide a prospective near-anchor check. Their public scores were 0.5963391483, 0.5963352712, and 0.5961947552: all worse than ID342 despite small local improvements. Other plausible additions—historical weather and air quality, sleep-boundary proxies, subject-state representations, and temporal convolution models—also failed the frozen local promotion gates and were not included.

Relative to ID178, ID342 reduces primary-lens loss most on S4 and Q1. However, the ID218 ablation shows that these target gains cannot be assigned wholly to the selected sensor families. The tiny ID218-to-ID221 changes on most targets further caution against modality-level

TABLE IV
Full-OOF alignment with public ranking.

Candidate set	n	Spearman	Pairwise
All scored	8	−0.167	0.536
Exclude ID287	7	0.250	0.714
Exclude ID287/329	6	1.000	1.000
June 23 slate	3	1.000	1.000

causal interpretation.

V. Negative Evidence and Reproducibility

Representative post-anchor routes were evaluated with the frozen gate. Three Q3 app-lexicon variants were publicly worse; sleep-boundary and temporal-CNN probes did not pass Q3/S3 stability checks. KMA ASOS produced only an unstable S4 change, while AirKorea 2024 promoted no target. A 450-configuration scan over circadian, GPS, phone/watch, proximity, and app-semantic matrices also produced no pass. These are sample-specific negative results, not evidence that the modalities are irrelevant.

Reproducibility was enforced at the artifact level. ID342 has a one-to-one mapping among its experiment record, out-of-fold matrix, test matrix, notebook snapshot, submission CSV, submission-log row, and DACON submission identifier 1467937. The stored CSV hash is 02f622e1e9c68257...e13b394; the official submission time is June 23, 2026, 23:30:37 KST. The final predictions reconstruct exactly as the 0.5/0.5 blend of ID178 and ID221. This lineage check prevents an unsubmitted local row from being reported as the final model.

The package excludes raw lifelogs and credentials. It includes the IEEE source, bibliography, metadata extracted from the source, lineage manifest, and commands for regenerating the audit reports. Random-convolution seeds and kernel counts are retained in the experiment metadata. Reproducing ID221 requires the ID218 OOF/test artifacts plus Table I; reproducing ID342 then requires ID178 and the 0.5 blend. Because selection is non-nested, 0.534000 is not independent validation.

VI. Discussion and Limitations

The result supports a narrow conclusion: a conservatively shrunk lineage ensemble with a domain-grouped similarity correction can improve a stable multimodal anchor. It does not establish that similarity caused the public gain or is superior to end-to-end temporal modeling. Only 10 participants are available, the search is not nested, and no public ablation isolates ID218 or the correction. Combined-covariate scaling is transductive and assumes the test batch is available. A deployment can instead fit preprocessing on historical training covariates; this changed ID221 OOF loss only from 0.532990 to 0.533015 in a train-only preprocessing check, but it is not independent test evidence and sequential distribution shifts may be larger. Moreover, the anchor and policy were developed after observing public scores, so the result is comparative competition evidence rather than an untouched test estimate.

The failed controls nevertheless reveal a useful operational pattern. When the sample is small and participant routines dominate, a dramatic local gain may indicate validation exploitation. Public-anchor shrinkage acts as regularization in prediction space, while failed-frontier alignment turns previous mistakes into explicit controls. Future work should evaluate the procedure with fully nested participant-aware selection, bootstrap uncertainty across person-days, and a private holdout from new participants. Self-supervised multimodal pretraining may then provide a genuinely independent source rather than another local recombination.

VII. Conclusion

We presented an anchor-constrained lineage ensemble with a small feature-similarity correction for seven sleep and stress targets. The complete model reduced DACON public log loss from 0.6014113009 to 0.5960832519 while keeping prediction movement small; the public contribution of the correction alone remains unidentified. The study shows why attribution ablations, explicit validation boundaries, prediction geometry, and negative controls matter in small-sample lifelog modeling.

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